

2.1.1 9's and 10's Complements

Subtraction of decimal numbers can be accomplished by the 9's and 10's complement methods similar to the 1's and 2's complement methods of binary. The 9's complement of a decimal number is obtained by subtracting each digit of that decimal number from 9. The 10's complement of a decimal number is obtained by adding a 1 to its 9's complement.

EXAMPLE 2.1 Find the 9's complement of the following decimal numbers.

(a) 3465

(b) 782.54

(c) 4526.075

Solution

$$\begin{array}{r} 9999 \\ - 3465 \\ \hline \end{array}$$

6534 (9's complement of 3465)

$$\begin{array}{r} 999.99 \\ - 782.54 \\ \hline \end{array}$$

217.45 (9's complement of 782.54)

$$\begin{array}{r} 9999.999 \\ - 4526.075 \\ \hline \end{array}$$

5473.924 (9's complement of 4526.075)

EXAMPLE 2.2 Find the 10's complement of the following decimal numbers.

(a) 4069

(b) 1056.074

Solution

$$\begin{array}{r} 9999 \\ - 4069 \\ \hline \end{array}$$

5930 (9's complement of 4069)

+ 1

Add 1

5931 (10's complement of 4069)

$$\begin{array}{r} 9999.999 \\ - 1056.074 \\ \hline \end{array}$$

8943.925 (9's complement of 1056.074)

+ 1

Add 1

8943.926 (10's complement of 1056.074)

2.1.2 9's Complement Method of Subtraction

To perform decimal subtraction using the 9's complement method, obtain the 9's complement of the subtrahend and add it to the minuend. Call this number the intermediate result. If there is a carry, it indicates that the answer is positive. Add the carry to the LSD of this result to get the answer. This is called end around carry. If there is no carry, it indicates that the answer is negative and the intermediate result is its 9's complement. Take the 9's complement of this result and place a negative sign in front to get the answer.

EXAMPLE 2.3 Subtract the following numbers using the 9's complement method.

(a) $745.81 - 436.62$ (b) $436.62 - 745.81$

Solution

$$\begin{array}{rcl}
 \text{(a)} & \begin{array}{r} 745.81 \\ - 436.62 \\ \hline 309.19 \end{array} & \Rightarrow \begin{array}{r} 745.81 \\ + 563.37 \text{ (9's complement of 436.62)} \\ \hline \textcircled{1} 309.18 \text{ (Intermediate result)} \\ \text{↪} \quad + 1 \text{ (End around carry)} \\ \hline 309.19 \text{ (Answer)} \end{array}
 \end{array}$$

The carry indicates that the answer is positive. So the answer is $+309.19$.

$$\begin{array}{rcl}
 \text{(b)} & \begin{array}{r} 436.62 \\ - 745.81 \\ \hline -309.19 \end{array} & \Rightarrow \begin{array}{r} 436.62 \\ + 254.18 \text{ (9's complement of 745.81)} \\ \hline 690.80 \text{ (Intermediate result with no carry)} \end{array}
 \end{array}$$

There is no carry indicating that the answer is negative. So, take the 9's complement of the intermediate result and put a minus sign. The 9's complement of 690.80 is 309.19.

Therefore, the answer is -309.19 .

2.1.3 10's Complement Method of Subtraction

To perform decimal subtraction using the 10's complement method, obtain the 10's complement of the subtrahend and add it to the minuend. If there is a carry, ignore it. The presence of the carry indicates that the answer is positive; the result obtained is itself the answer. If there is no carry, it indicates that the answer is negative and the result obtained is its 10's complement. Obtain the 10's complement of the result and place a negative sign in front to get the answer.

EXAMPLE 2.4 Subtract the following numbers using the 10's complement method.

(a) $2928.54 - 416.73$ (b) $416.73 - 2928.54$

Solution

$$\begin{array}{rcl}
 \text{(a)} & \begin{array}{r} 2928.54 \\ - 0416.73 \\ \hline 2511.81 \end{array} & \Rightarrow \begin{array}{r} 2928.54 \\ + 9583.27 \text{ (10's complement of 416.73)} \\ \hline \textcircled{1} 2511.81 \text{ (Ignore the carry)} \end{array}
 \end{array}$$

There is a carry indicating that the answer is positive. Ignore the carry.

The answer is 2511.81.

$$\begin{array}{r}
 \text{(b)} \quad 0416.73 \\
 - 2928.54 \\
 \hline
 -2511.81
 \end{array}
 \Rightarrow
 \begin{array}{r}
 0416.73 \\
 + 7071.46 \quad (\text{10's complement of } 2928.54) \\
 \hline
 7488.19 \quad (\text{No carry})
 \end{array}$$

There is no carry indicating that the answer is negative. So, take the 10's complement of the intermediate result and put a minus sign. The 10's complement of 7488.19 is 2511.81. Therefore, the answer is -2511.81 .